

Quenverell Computational Ltd, a Tarnovell company

Stress-kinase ligands: crystallography-based docking and pocket-water assessment

Final client report

Engagement: QV.25.086

Client: Drelvessa Therapeutics SAS

Date: 13/11/2025

Author: Dr T. Zawiernicki, Head of Bioinformatics

1 Question

Drelvessa's chemists asked where their stress-kinase series could be extended without losing the hinge interaction, and whether conserved water molecules in the ATP pocket should be displaced or kept.

2 Data provenance

- Three client-supplied X-ray crystal structures of the kinase domain with bound ligands (resolution between 1.8 and 2.3 Å)
- Electron density maps reviewed for ligand and water placement before use
- Client compound set of current analogues and proposed designs (SMILES), checked for tautomer and protonation states

Data acceptance: one structure was excluded from scoring because the ligand density did not support the deposited pose; it was used only for visual comparison.

3 Methods

- Protein preparation from the crystal structures with conserved waters retained or removed in separate runs
- Docking with hinge hydrogen-bond constraints; pose clustering and visual review with the client's chemists
- Pocket-water assessment: waters seen in all three structures treated as conserved; their displacement scored separately
- Ranking by consensus across the two retained crystal structures

Docking scores rank poses within this pocket only; they are not binding affinities and are not compared across chemotypes.

4 Findings

- The hinge-binding pose of the series is reproduced in both retained crystal structures.
- Two conserved waters bridge the ligand to the back pocket; designs that displace them lost the consensus pose and are not recommended.
- A solvent-facing vector from the 5-position is open in both structures and is predicted to accommodate polar substituents.
- Designs extending towards the glycine-rich loop were ranked inconsistently between structures and are flagged as uncertain.

5 Recommendation

Prioritise designs using the solvent-facing 5-position vector with small polar groups, keep the conserved waters, and confirm binding in the client's biochemical assay before any further structure-based work. A new co-crystal structure with a representative analogue could help test the proposed pose near the glycine-rich loop.

6 Limitations

Conclusions rest on two accepted crystal structures of one conformational state. Induced-fit effects and protein flexibility were not modelled beyond side-chain refinement.

Name	Role	Signature	Date
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Appendix A Provenance and settings

Table A1 Provenance

Item	Value
Structures received	DRV-XS-01, DRV-XS-02, DRV-XS-03 (client internal IDs)
Structures used for scoring	DRV-XS-01, DRV-XS-03
Excluded	DRV-XS-02 (ligand density does not support deposited pose; visual comparison only)
Conserved waters	W1 (hinge-adjacent), W2 (back pocket); present in all three structures
Docking workflow	Quenverell docking pipeline v4.2, hinge H-bond constraint, 20 poses per ligand
Protonation	pH 7.4 states with tautomer enumeration
Designs assessed	DRV-D-01 to DRV-D-24 (client design list of 08/09/2025)
Analysis notebook	QV.25.086/notebooks/docking_consensus (versioned)

Table A2 Design ranking (qualitative)

Design group	Designs	Assessment
5-position polar	DRV-D-03, D-07, D-11, D-12, D-19	Recommended: consistent pose in both structures
5-position lipophilic	DRV-D-04, D-08	Possible; lower consensus
Water-displacing	DRV-D-01, D-02, D-15, D-16	Not recommended: pose lost
Glycine-loop extension	DRV-D-20 to D-24	Uncertain: ranks differ between structures